

SUVANWITA DAS

📍 Suvanwita 🏠 Suvanwita Das ✉️ dsuvanwita@gmail.com 📞 +91-6291212208

EDUCATION

- **Indian Institute of Information Technology, Allahabad** Prayagraj, Uttar Pradesh
B.Tech- Information Technology - CGPA: 8.92 August 2024 - Present
- **Central Model School** Kalyani, West Bengal
Class 12 CBSE - Percentage - 95.4% 2024
- **Julien Day School** Kalyani, West Bengal
Class 10 ICSE - Percentage - 98.8% 2022

EXPERIENCE

- **University of Missouri** Jan 2026 – Feb 2026
Software Development Engineer Intern Remote
 - Designed and implemented a graph-based retrieval system using LangChain, LangGraph, and Neo4j to support workflow-aware genomic analysis assistance.
 - Developed backend validation pipelines and data processing workflows for large-scale genomic datasets, improving data quality and reducing manual verification effort.
 - Enhanced scientific visualization modules through custom grouping and analysis features for volcano plot generation.

SKILLS

Languages: C++, Python, Java, JavaScript, TypeScript, Rust, C, Bash/Shell
Backend & Systems: Node.js, Express, FastAPI, Redis, Kafka, BullMQ, PostgreSQL, MongoDB, MySQL, REST APIs, WebSockets, Socket.io, Jest, Supertest
Frontend: React.js, Next.js, HTML, CSS, Tailwind CSS, Zustand, Framer Motion
AI/ML: Scikit-Learn, OpenCV, LangChain, LangGraph, NumPy, Pandas, Matplotlib, Seaborn
Developer Tools: Git, GitHub, Linux, Docker, Kubernetes, GitHub Actions, Postman
Core Concepts: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

PROJECTS

- **SwiftCache (C++, CMake)** 🔗 GitHub
 - Developed a Redis-inspired **in-memory datastore** in C++17 featuring a **multithreaded TCP server** that parses the RESP protocol and inline TCP requests.
 - Engineered **thread-safe shared data access** using mutexes, supporting core cache structures, logical databases, TTL expiration, and configurable eviction policies like LRU.
 - Integrated durability via **AOF logging** and **binary snapshot persistence** with background worker threads, alongside client management and real-time server metrics.
- **EventPulse (Next.js, Node.js, Express, PostgreSQL, Redis, Kafka, Socket.io)** 🔗 GitHub
 - Built a campus event operations platform supporting capacity-safe registrations, automated waitlists, QR-based entry, and real-time crowd analytics.
 - Secured the API layer using **Zod** for robust request validation and **CASL** for role-based authorization policies. Architected an event-driven backend leveraging **Kafka** asynchronous processing, **PostgreSQL** transactions, and **Redis** locking to ensure strict data consistency, successfully eliminating race conditions during high-concurrency workloads of up to **1,000 concurrent requests**.
 - Reduced analytics query latency by **45.3%** (from 56.8ms to 31.1ms) using **Fenwick Tree** aggregation and improved autocomplete speed by **98.6%** (from 3,106.5ms to 42.9ms) via **Trie-based prefix lookups**. Offloaded live-counter queries to Redis, decreasing query latency by over **99.9%** (from 6,772.7ms to 1.2ms).
- **SheCare (Next.js, Node.js, Express, MongoDB, Python, FastAPI, Scikit-learn, Redis, Kafka, BullMQ)** 🔗 Github
 - Developed a full-stack women's healthcare platform featuring cycle tracking, PCOS assessment, appointment scheduling, and medical record management.
 - Built an AI-powered health intelligence layer using **FastAPI microservices**, with **94% accuracy** (90% F1-score) for PCOS prediction and **95% accuracy** for cycle irregularity forecasting using **Random Forest** models. Engineered TF-IDF recommendation systems and optimized backend performance using **Redis** caching, **BullMQ** workers and **Kafka** event streaming.

ACHIEVEMENTS

- Among Top 200 scholars selected for **Flipkart Girls Wanna Code 7.0 Top Scholars Cohort** all over India
- Achieved **Knight on Leetcode** 🔗, **3 star on Codechef** 🔗 and **Pupil on Codeforces** 🔗
- Ranked 14 among 150+ participants in Code-X-Culture'26, annual coding contest, IIITA
- Secured Rank 12 in OpenCode 2024, the open-source competition conducted by GeekHaven IIITA
- Attained Rank 9 in the Out Of Context Hackathon at Effervescence IIITA

POSITIONS OF RESPONSIBILITY

- **Member — FOSS Wing, Geekhaven:** Conducted hands-on workshops for 100+ first-year students. Maintained the official OpenCode website; Served as a **Mentor for OpenCode'25** - mentored **70+ contributors** for the **CareerCraft** repository, systematically tracking **40+ issues** and successfully reviewing and merging **150+ pull requests**.
- **Mentor — GirlScript Summer of Code (GSSoC) 2026:** Guided global open-source contributors on technical workflows, conducted code reviews, and managed pull requests.